

Supplementary Material: ControlledRAG: A Seven-Axis Reporting Standard for RAG Faithfulness Claims

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1 Pre-Specification

The matched-context, scaled, metric-fragility, threshold-transfer, noise, cost-aware-baseline, and human-calibration audits were pre-specified on 2026-04-26 before any results were inspected. The

human-calibration design specified two arms: (i) a stratified $n \approx 99$ two-pass adjudicated calibration on the typical row distribution of the 7,500 Mistral generations, scored as raw inter-pass agreement, Cohen’s κ , and Spearman correlations of each scorer against the adjudicated label; (ii) a targeted 100-row scorer-disagreement batch constructed from the same 7,500 rows and designed to stress-test scorer conflict using two independent raters plus an adjudication step. Both arms are now completed: §12 reports arm (i) at $n = 99$ adjudicated items, and §13 reports arm (ii) at $n = 100$ two-rater adjudicated items. The pre-specification also fixed: the matched-pair construction (similarity gap ≤ 0.02 , CCS gap ≥ 0.20); the paired Wilcoxon test direction (HIGH>LOW); the 10,000-resample bootstrap for the CI on the mean paired difference; the Wilson 95% CI on hallucination rates; the seed list {41, 42, 43, 44, 45}; the per-seed paired Wilcoxon as the primary significance test in the scaled audit; the three faithfulness scorers (DeBERTa proxy, **roberta-large-mnli** proxy, RAGAS-style) plus the fixed-context rescore protocol; the τ -sweep range; and the 200-query head-to-head sample size. The four follow-up arms were added after the pre-specified runs as post-hoc robustness checks: answer-span/control diagnostic, stronger-retriever sanity check, Qwen2.5 second-generator replication, and long-form stress test. We disclose which experiments are pre-specified (`audit_log.md` sections under “Pre-Specified”) versus follow-ups (`audit_log.md` sections under “Follow-Up”) in the file `audit_log.md` shipped at the top level of the artifact.

2 Scaled Audit: Per-Seed Detail

The scaled audit on SQuAD/Mistral runs $n = 2,500$ examples per condition for each of five seeds; pooled means appear in the main paper. Table 1 reports per-seed paired Wilcoxon p -values, mean paired differences, and Cohen’s d_z for each contrast.

Table 1: Per-seed paired contrasts on SQuAD/Mistral, $n = 500$ pairs per seed. Significance is paired Wilcoxon at $\alpha = 0.05$.

Seed	Contrast	Mean diff	Wilcoxon p	Cohen’s d_z
41	baseline–v1	+0.0107	0.083	+0.072
42	baseline–v1	+0.0063	0.234	+0.043
43	baseline–v1	+0.0078	0.174	+0.055
44	baseline–v1	+0.0204	0.008	+0.134
45	baseline–v1	+0.0073	0.203	+0.048
41	v2–v1	+0.0132	0.025	+0.088
42	v2–v1	+0.0030	0.504	+0.021
43	v2–v1	+0.0059	0.215	+0.042
44	v2–v1	+0.0191	0.008	+0.126
45	v2–v1	+0.0125	0.036	+0.081

Bootstrap CIs on the pooled mean paired difference for baseline–v1 include zero, and only 1 of 5 seeds reaches significance for that contrast. The v2–v1 contrast reaches significance in 3 of 5 seeds, but the magnitude is small (per-seed mean diff 0.003–0.019). Seed-level uncertainty is treated as a ControlledRAG reporting axis, not a result to average away.

3 Answer-Span/Control Diagnostic Details

The answer-span detector lowercases text, strips punctuation, normalises whitespace, and checks whether any normalised gold-answer alias appears in the retrieved context for the audited RAG

setting [Lewis et al., 2020]. It uses no external model. The resulting control rows and model summaries are written under the Fix 12 data and results directories.

Table 2: Matched-context span and outcome summary.

Group	n	span present	faithfulness	hallucination	CCS
all	400	0.220	0.637	0.128	0.404
HIGH-CCS	200	0.170	0.636	0.165	0.670
LOW-CCS	200	0.270	0.639	0.090	0.137
span absent	312	0.000	0.628	0.119	0.408
span present	88	1.000	0.670	0.159	0.390

Table 3: Full span/control model comparison. CIs are bootstrap 95% intervals from the script output.

Predictors	R^2	R^2 CI	AUROC	AUROC CI
answer span	0.019	[0.001, 0.062]	0.531	[0.502, 0.599]
mean similarity	0.002	[0.000, 0.023]	0.586	[0.495, 0.677]
CCS	0.003	[0.000, 0.022]	0.639	[0.564, 0.712]
span + similarity	0.024	[0.004, 0.071]	0.590	[0.518, 0.687]
span + CCS	0.022	[0.003, 0.068]	0.631	[0.572, 0.716]
span + similarity + CCS	0.025	[0.006, 0.075]	0.643	[0.589, 0.734]

The headline reading is reported in the main paper: span presence is the strongest single predictor of continuous faithfulness, CCS is the strongest single predictor of hallucination AUROC, and the joint feature set has the highest score on both. The R^2 values are uniformly small in absolute terms; the relative ranking is the load-bearing signal. Span presence is a weak lexical proxy, not a gold support variable: retrieved text can contain the answer string while still failing to support the generated answer. The mean-score and thresholded hallucination views also inspect different parts of the score distribution.

Table 4: Matched HIGH/LOW CCS faithfulness distribution.

Group	n	mean	p10	p25	p50	p75	p90
HIGH-CCS	200	0.636	0.476	0.553	0.638	0.713	0.795
LOW-CCS	200	0.639	0.512	0.569	0.648	0.715	0.769

Table 5: Matched HIGH/LOW CCS hallucination rates as the score threshold varies.

Group	$\theta = 0.3$	$\theta = 0.4$	$\theta = 0.5$	$\theta = 0.6$	$\theta = 0.7$
HIGH-CCS	0.0%	3.0%	16.5%	38.5%	71.0%
LOW-CCS	1.0%	3.0%	9.0%	38.5%	70.0%

These tables explain why the matched mean contrast can be null while a specific thresholded hallucination contrast differs: the distributions are close in the middle, and the observed HIGH/LOW difference is threshold-sensitive rather than a clean directional CCS effect.

4 Stronger Retriever Sanity Check

The retriever sanity check has two parts. First, the matched contexts from the main-paper audit are re-encoded with MiniLM, BGE-large, and E5-large; this is a fixed-context diagnostic, not a fresh generation run. Second, the completed multi-retriever generation artifact is copied into `data/revision/fix_13/` and summarised. The fixed generations are unchanged across embedders, so the matched HIGH–LOW faithfulness contrast inherits the matched-context null from the main paper and is not re-tested here.

Table 6: Fixed-context re-encoding of the matched HIGH/LOW CCS pairs. The HIGH>LOW CCS ordering is preserved across all three embedders.

Embedder	pairs	CCS high	CCS low	high–low	sim gap
MiniLM	200	0.670	0.137	0.533	0.006
BGE-large	200	0.739	0.491	0.249	0.032
E5-large	200	0.830	0.702	0.127	0.015

Table 7: Paradox magnitude by embedding model from the completed multi-retriever generation artifact ($n = 30$ per cell, original pilot sampling protocol re-run under different embedders). The SQuAD column shows that the paradox direction depends on retriever; PubMedQA preserves a positive paradox across all four embedders. *These rows are not directly comparable to the $n = 2,500$ scaled-audit number in main paper §6 (baseline–v1 = 0.011): that table uses the audit-construction sampling and a single MiniLM retriever, whereas this table uses the small- n pilot sampling and varies the embedder. Cross-table comparison should be same-row across embedders within this table, not this table vs. the main-paper scaled-audit table.*

Dataset	embedder	baseline–v1	v2–v1
SQuAD	MiniLM	+0.075	+0.099
SQuAD	BGE-large	+0.032	+0.025
SQuAD	E5-large	−0.011	+0.008
SQuAD	GTE-large	−0.001	−0.006
PubMedQA	MiniLM	+0.024	+0.019
PubMedQA	BGE-large	+0.033	+0.028
PubMedQA	E5-large	+0.053	+0.039
PubMedQA	GTE-large	+0.006	+0.002

The safe interpretation is retriever sensitivity, not universal robustness. Stronger embeddings change the retrieval surface, sometimes in a direction that flips the SQuAD paradox sign. ControlledRAG asks for retriever identity to be reported as a routine audit axis; the supplement here is the operational example.

5 Second-Generator Probe (Qwen2.5)

The second-generator audit samples $n = 100$ fixed contexts per condition from the matched-context construction and generates answers with Qwen2.5 via local Ollama. Those fixed generations are scored with the DeBERTa proxy, the `roberta-large-mnli` proxy, and a local RAGAS-style judge (Mistral via Ollama).

The scorer-disagreement ordering is preserved under this compact second-generator replication. The claim is limited to this audited Qwen2.5 cell; ControlledRAG explicitly asks for a broader

Table 8: Qwen2.5 fixed-generation metric means and raw score contrasts.

Metric	baseline	HCPC-v1	HCPC-v2	baseline-v1	v2-v1
DeBERTa	0.645	0.631	0.644	+0.015	+0.014
roberta-large-MNLI	0.362	0.316	0.368	+0.046	+0.051
RAGAS-style	0.806	0.672	0.814	+0.134	+0.142

Table 9: Metric correlations on the Qwen2.5 fixed-generation rows ($n = 300$).

Metric pair	Pearson r	Spearman ρ
DeBERTa vs. roberta-large-MNLI	0.380	0.429
DeBERTa vs. RAGAS-style	0.280	0.318
roberta-large-MNLI vs. RAGAS-style	0.721	0.668

generator panel before any universal-generator claim is made.

6 Scorer Implementation and Standardized Contrasts

The frozen artifact uses two legacy NLI-style zero-shot label proxies and one context-conditioned RAGAS-style judge. The DeBERTa proxy (`cross-encoder/nli-deberta-v3-base`) and RoBERTa proxy (`roberta-large-mnli`) score each generated answer sentence as the sequence to be classified against the labels entailment/neutral/contradiction with `hypothesis_template="{}`". The retrieved context is retained in the per-row data and used by the RAGAS-style judge, but it is not consumed by those two legacy zero-shot proxy calls. This disclosure is part of the scorer audit: model identity, input formatting, and prompt format are not implementation trivia.

Because the three scorers are on different native scales, the main paper reports raw contrasts as score-scale-dependent contrasts and adds a standardized Mistral analysis. Table 10 z-scores each scorer over the same 7,500 fixed-generation rows and also reports percentile-rank-normalized contrasts. CIs are paired bootstrap intervals over matched query/seed keys; 2,499 paired keys are available after one duplicated key is collapsed by the pivot.

Table 10: Standardized fixed-generation scorer-fragility contrasts on Mistral rows. Positive values mean the left condition scores higher than HCPC-v1.

Scorer	Contrast	raw diff		z diff (CI)	rank diff (CI)
DeBERTa proxy	baseline-v1	0.010	0.071	[0.032, 0.110]	0.022 [0.010, 0.033]
DeBERTa proxy	v2-v1	0.011	0.073	[0.034, 0.112]	0.022 [0.011, 0.034]
roberta-large-MNLI proxy	baseline-v1	0.032	0.231	[0.191, 0.271]	0.062 [0.051, 0.074]
roberta-large-MNLI proxy	v2-v1	0.032	0.236	[0.199, 0.275]	0.063 [0.052, 0.075]
RAGAS-style judge	baseline-v1	0.139	0.337	[0.298, 0.375]	0.085 [0.075, 0.095]
RAGAS-style judge	v2-v1	0.138	0.332	[0.294, 0.370]	0.083 [0.073, 0.092]

The standardized table preserves the scorer-fragility conclusion without treating raw scores on different scales as directly comparable causal effects.

7 Context-Conditioned NLI Rescoring (Subset)

The frozen artifact’s DeBERTa-v3 and `roberta-large-mnli` faithfulness columns are legacy zero-shot label proxies: the answer sentence is the sequence classified against the labels entailment / neutral / contradiction with `hypothesis_template="{}`", and the retrieved context is not consumed by those calls. To make scorer input format an empirically observable audit axis (not just a disclosure axis), we rescore a balanced subset of the same fixed-generation rows under a proper context-conditioned NLI calling convention:

premise = retrieved context (truncated to 2048 chars), hypothesis = generated answer sentence,

followed by softmax over {contradiction, neutral, entailment} and a mean-over-sentences aggregation matching the legacy proxy aggregation. The rescoring script is `scripts/run_context_conditioned_nli_scoring`.

Subset. $n = 200$ paired queries per condition (baseline, HCPC-v1, HCPC-v2), so $n = 600$ rows total. Subset selection is deterministic (first 200 rows by appearance under the original `fix_03` ordering), so the subset is balanced across conditions but is not stratified by question domain. Rescoring is performed with both `cross-encoder/nli-deberta-v3-base` and `roberta-large-mnli` on Apple-silicon MPS without paid APIs.

Table 11: Baseline–HCPC-v1 paired contrasts on the same fixed generations under five scorer input formats. Bootstrap 95% CIs are 10,000-resample paired bootstraps over matched query keys. The two legacy zero-shot proxies (rows 1–2) and the two corresponding context-conditioned NLI rescorings (rows 3–4) use the same backbone NLI weights and the same fixed answers; only the calling convention changes.

Scorer	Baseline–HCPC-v1 contrast	95% CI
Legacy DeBERTa-v3 proxy (no context)	+0.017	[−0.003, +0.037]
Legacy roberta-large-mnli proxy (no context)	+0.044	[+0.026, +0.063]
Context-conditioned DeBERTa-v3 NLI	−0.279	[−0.340, −0.219]
Context-conditioned roberta-large-mnli NLI	−0.081	[−0.122, −0.043]
Context-conditioned RAGAS-style judge	+0.130	[+0.075, +0.186]

Reading. For both DeBERTa-v3 and **roberta-large-mnli**, switching from the legacy zero-shot label proxy to a context-conditioned NLI calling convention on the same backbone weights changes the sign of the reported baseline–HCPC-v1 contrast on the same fixed generations. The legacy and context-conditioned bootstrap CIs are disjoint for both models. The RAGAS-style judge, which is context-conditioned by construction, agrees in sign with the legacy proxies but disagrees in magnitude. We do not claim a uniformly “correct” scorer; the table is the audit point: scorer input format is a real movable quantity in the headline number.

Alignment with the $n = 99$ calibration. On the $n = 99$ adjudicated calibration items, switching the same DeBERTa-v3 and **roberta-large-mnli** backbones to the context-conditioned calling convention does *not* improve Spearman alignment with the adjudicated labels: legacy DeBERTa proxy $\rho = +0.157$ vs. context-conditioned DeBERTa $\rho = +0.120$; legacy **roberta-large-mnli** proxy $\rho = +0.393$ vs. context-conditioned $\rho = +0.224$ (CIs in `n99_alignment_spearmans.csv`). The result is consistent with NLI models pre-trained on short premise/hypothesis pairs being sensitive to long-context input format. This is a calibration finding, not a recommendation; the contribution is that scorer input format must be reported and audited, not that any one input format is preferred.

Subset is not the full table. This is a 600-row subset, not a full 7,500-row rescoring. The scaled audit raw table in the main paper still uses the full 7,500-row legacy proxy values; the standardised z-/rank-contrast table (§6) also uses the legacy proxies. The context-conditioned subset is a focused empirical demonstration of the input-format axis, not a replacement for the legacy table.

Source `files. scripts/run_context_conditioned_nli_scoring.py,`
`results/revision/context_conditioned_nli/per_query_n600.csv,`
`results/revision/context_conditioned_nli/contrasts_n600.csv,`
`results/revision/context_conditioned_nli/n99_with_ctx_scores.csv,`
`results/revision/context_conditioned_nli/n99_alignment_spearmans.csv.`

8 Long-Form/Synthesis Stress Test

The long-form stress test packages an existing QASPER/MS-MARCO run with 40 distinct long-form questions. It is included only as an exploratory scope probe, not as broad generalization evidence for the short-answer audit.

Table 12: Long-form stress-test condition means.

Dataset	condition	span faith	claim faith	unsupported	long halluc.
MS-MARCO	baseline	0.645	0.646	0.224	0.400
MS-MARCO	HCPC-v1	0.678	0.679	0.075	0.050
MS-MARCO	HCPC-v2	0.674	0.674	0.158	0.250
QASPER	baseline	0.649	0.649	0.158	0.300
QASPER	HCPC-v1	0.647	0.647	0.117	0.200
QASPER	HCPC-v2	0.648	0.651	0.129	0.250

The short-answer paradox does not generalise cleanly here: MS-MARCO has a negative span-faithfulness paradox drop (-0.033), and QASPER is near zero ($+0.002$). The long-form result is reported as a scope limit rather than as broad long-form evidence for ControlledRAG.

9 Self-RAG: Harness-Mismatched Appendix Baseline

We include Self-RAG [Asai et al., 2024] only as an appendix baseline because the available Self-RAG harness mismatches the audit’s fixed-context protocol along three dimensions: (i) Self-RAG’s generator emits its own retrieval and reflection tokens, so the “fixed retrieved context” assumption used by the matched-context and metric-fragility audits is not directly satisfied; (ii) Self-RAG’s recommended generator is a fine-tuned Llama-2-7B variant that does not coincide with the audited Mistral-7B / Qwen2.5-7B family and that was OOM-killed on the available T4 instance for the PubMedQA cell, so the partial SQuAD numbers we have are not directly comparable to the cost-aware head-to-head; and (iii) the recommended Self-RAG threshold differs from the threshold-transfer sweep used elsewhere. We report this disclosure here so that future ControlledRAG-style audits can budget for a matched Self-RAG harness; we explicitly do not generalize from these mismatched runs to Self-RAG overall.

10 Coherence vs. Generic Retrieval Noise

The noise audit uses the matched-context construction as a backbone and replaces 1/3, 2/3, or 3/3 of the retrieved chunks with one of two noise types: random off-topic passages and coherence-preserving uninformative passages drawn from the same topic distribution that do not contain the answer.

Table 13: Faithfulness and similarity slopes per noise rate.

Condition	faith slope (per noise rate)	sim slope (per noise rate)
random noise	-0.069	-0.481
coherent uninformative noise	-0.043	-0.113
HCPC-v1 refinement (reference)	—	—

The two noise types produce distinguishable faithfulness slopes at matched substitution rate; we do not claim the gap is large in absolute terms. The right reading is that coherence and similarity are correlated but not identical.

11 Threshold Transfer: Full Sweep

Naming. Tables in this section use the prior-work names HCPC-v1 (= aggressive-refinement) and HCPC-v2 (= gated-refinement); see main paper §3.

The main paper reports a per-tune-dataset summary (diagonal recovery, off-diagonal mean, gap) at each tune dataset’s selected τ^* . The full 5×5 per-cell transfer matrix at each tune dataset’s τ^* is reproduced below; the per- τ sweep over $\{0.3, 0.4, 0.5, 0.6, 0.7\}$ for each dataset is in the artifact’s `results/revision/fix_04/` directory. Recovery is the dimensionless normalized contrast

$$R_\tau(d) = \frac{\text{faith}_{\text{gate},\tau}(d) - \text{faith}_{\text{HCPC-v1}}(d)}{\text{faith}_{\text{baseline}}(d) - \text{faith}_{\text{HCPC-v1}}(d)}.$$

It is useful for asking whether a tuned threshold transports, but it is not a direct utility measure: when the denominator is small or negative, the sign can be counterintuitive and values can exceed 1.

Table 14: Full 5×5 threshold-transfer matrix. Each row gives the normalized recovery ratio of the τ tuned on the row dataset, evaluated on each of the five column datasets. Positive values are not always “better” when the baseline–v1 denominator is negative. Rows can be byte-identical across the eval columns when two tune datasets pick the same τ^* : tune-SQuAD and tune-HotpotQA both pick $\tau^* = 0.3$, and tune-PubMedQA and tune-TriviaQA both pick $\tau^* = 0.4$, so the off-diagonal columns coincide. This is a faithful consequence of the experiment (off-diagonal recovery only depends on the eval dataset and the selected τ , not on which tune dataset selected it), not a pipeline bug.

τ tuned on	τ^*	SQuAD	PubMedQA	HotpotQA	NaturalQs	TriviaQA
PubMedQA	0.4	+0.489	+1.452	−0.291	+1.140	+0.464
NaturalQs	0.5	+0.327	+0.664	−0.471	+1.401	+0.323
SQuAD	0.3	+0.823	−1.714	−0.112	+0.865	+0.398
TriviaQA	0.4	+0.489	+1.452	−0.291	+1.140	+0.464
HotpotQA	0.3	+0.823	−1.714	−0.112	+0.865	+0.398

Per- τ recovery curves and confidence diagnostics for the τ -tuning step are included as machine-readable CSVs in the same results directory. The transfer matrix itself does not include bootstrap CIs, so we use it as a descriptive transport diagnostic.

12 Completed Annotation Calibration ($n = 99$)

The $n = 99$ annotation calibration set is the typical-row arm of the human-calibration pre-specification. Items were sampled from the 7,500 Mistral fixed-generation rows so that the baseline, HCPC-v1, and HCPC-v2 conditions are each represented and so that the score range of each scorer is covered. Two independent annotation passes were performed using the three-class label scheme `supported` / `partially_supported` / `unsupported`; disagreements were adjudicated and the adjudicated label set is the final reference for scorer-alignment computations.

Adjudicated label distribution. The adjudicated label distribution is 79 `supported`, 16 `partially_supported`, 4 `unsupported` (totals 99). Raw inter-pass agreement on the three-class label is 0.919 (91 of 99 items exact-match), and Cohen’s $\kappa = 0.774$.

Table 15: Annotation calibration agreement and per-scorer alignment on $n = 99$ adjudicated examples. Spearman, Pearson, and Kendall correlations are between the ordered adjudicated label (`unsupported` < `partially_supported` < `supported`) and each scorer’s continuous output on the same items. Bootstrap 95% CIs are paired bootstrap intervals; the three Spearman CIs overlap, so the per-scorer ranking is suggestive only.

Scorer / quantity	Spearman ρ	Pearson r	Kendall τ_b	Spearman CI
Legacy DeBERTa proxy	0.103	0.110	0.083	$[-0.080, +0.283]$
Legacy <code>roberta-large-mnli</code> proxy	0.394	0.404	0.325	$[+0.222, +0.543]$
Context-conditioned RAGAS-style judge	0.549	0.450	0.527	$[+0.380, +0.707]$
Cohen’s κ between the two annotation passes: 0.774				
Raw exact-match agreement: 0.919 (91 of 99)				

Reading. The Spearman ordering matches the metric-fragility ordering in the main paper (RAGAS-style judge > `roberta-large-mnli` proxy > DeBERTa proxy), but two facts keep this calibration short of a settled scorer endorsement: (i) the three Spearman bootstrap CIs overlap, and (ii) $n = 99$ is calibration sample size, not population sample size. The legacy DeBERTa proxy’s CI overlaps zero, while the other two CIs do not—this is the strongest single line that can be drawn under this calibration.

Source files. The verification script `scripts/verify_human_eval.py` re-computes the table values from the combined two-pass annotation file `human_eval_final/n99_calibration/human_eval_adjudicated.csv` (carrying `rater_a_label`, `rater_b_label`, `adjudicated_label`, `adjudication_notes`) and emits the bootstrap-CI summary above. The label distribution, agreement summary, correlation table, bootstrap intervals, verification trace, and per-row file with retrieved-context column appear in the same `human_eval_final/n99_calibration/` directory under `human_eval_label_distribution.csv`, `human_eval_summary.csv`, `human_eval_correlations.csv`, `bootstrap_correlation_cis.csv`, `human_eval_verification.csv`, and `human_eval_n99_with_context.csv`.

13 Completed Targeted Scorer-Disagreement Evaluation ($n = 100$)

The targeted scorer-disagreement evaluation is the adversarial arm of the human-calibration pre-specification. It is *separate* from the $n = 99$ typical-row calibration in §12: items are constructed to maximise scorer disagreement rather than to estimate population-level faithfulness. Both raters labelled all 100 items and adjudication resolved the 12 rater-rater disagreements.

Batch construction. Rows are sampled from the 7,500 Mistral generations in the scaled audit. We compute the absolute disagreement between the legacy DeBERTa proxy and the context-conditioned RAGAS-style score and stratify the batch across three disagreement types:

1. **DeBERTa high / RAGAS-style low.**
2. **RAGAS-style high / DeBERTa low.**
3. **Near-threshold disagreement** around the 0.5 decision boundary.

The batch is also balanced across baseline / HCPC-v1 / HCPC-v2. Items already used in the $n = 99$ calibration are excluded.

Annotation protocol. Two independent raters assigned one of `faithful`, `hallucinated`, or `unclear` using only the retrieved context, and an adjudication step resolved disagreements. The protocol artifacts shipped with the review package are:

- `human_eval_final/n100_disagreement/annotation_batch_disagreement_100.csv`: the merged batch with scorer scores, rater 1, rater 2, and adjudicated labels.
- `human_eval_final/n100_disagreement/human_eval_rater1.csv` and `human_eval_rater2.csv`: independent rater files used during labelling.
- `human_eval_final/n100_disagreement/human_eval_adjudicated.csv`: per-row file with rater 1 label, rater 2 label, adjudicated label, adjudicated confidence, and adjudication notes; all 12 rater-rater disagreements carry meaningful adjudication notes.
- `human_eval_final/n100_disagreement/human_eval_adjudication_template.csv`: the reconciliation template used during adjudication.

Agreement and label distribution. The two raters labelled all 100 items (100/100 for each rater); the adjudicated label set covers all 100. Raw inter-rater agreement is 0.88 (88 of 100 exact-match), Cohen’s $\kappa = 0.721$. The rater 1 distribution is 74 `faithful` / 26 `hallucinated` / 0 `unclear`; the rater 2 distribution is 70 / 20 / 10 (rater 2 used the `unclear` option, rater 1 did not). The adjudicated distribution is 74 `faithful` / 26 `hallucinated` / 0 `unclear`.

Scorer alignment with adjudicated labels. Table 16 reports per-scorer correlation, AUROC, and AUPRC against the adjudicated `faithful` / `hallucinated` label, with 1,000-resample bootstrap 95% CIs from `scripts/analyze_human_disagreement_labels.py`.

Table 16: Two-rater adjudicated $n = 100$ targeted scorer-disagreement evaluation. The slice is constructed to be adversarial for any single scorer; the ranking here is *not* the typical-row ranking from the $n = 99$ calibration. Raw inter-rater agreement is 0.88; Cohen’s $\kappa = 0.721$. The adjudicated distribution is 74 `faithful` / 26 `hallucinated` / 0 `unclear`.

Scorer	Spearman ρ	Pearson r	AUROC	AUROC CI	AUPRC	AUPRC CI
Legacy DeBERTa proxy	-0.156	-0.188	0.398	[0.273, 0.514]	0.765	[0.670, 0.852]
Legacy <code>roberta-large-mnli</code> proxy	+0.446	+0.380	0.794	[0.699, 0.879]	0.929	[0.881, 0.963]
Context-conditioned RAGAS-style judge	+0.384	+0.366	0.718	[0.637, 0.793]	0.859	[0.793, 0.914]

Reading. On this targeted disagreement-focused slice, `roberta-large-mnli` aligns best with adjudicated labels by all three measures (Spearman $\rho = 0.446$, AUROC 0.794, AUPRC 0.929); RAGAS-style is second; the legacy DeBERTa proxy is weakly negative on correlation and below chance on AUROC. This reverses the typical-row ordering from the $n = 99$ calibration, where RAGAS-style ranked strongest. The combination supports the ControlledRAG claim that scorer ranking is evaluation-slice dependent, not a global ordering.

Caveat. The $n = 100$ batch is constructed to be adversarial for any single scorer (rows where the legacy DeBERTa proxy and the RAGAS-style judge disagree most). It is therefore not a population-level faithfulness estimate and not a substitute for the $n = 99$ random-sample calibration. The two slices answer two different audit questions, and ControlledRAG asks for both to be reported.

Source files. All under human_eval_final/n100_disagreement/:
 annotation_batch_disagreement_100.csv, human_eval_rater1.csv, human_eval_rater2.csv,
 human_eval_adjudicated.csv (rater 1, rater 2, adjudicated, notes),
 human_disagreement_summary.csv, human_disagreement_label_distribution.csv,
 human_disagreement_agreement.csv, human_disagreement_correlations.csv,
 human_disagreement_aurocauprc.csv, human_disagreement_bootstrap_cis.csv.

14 Source Trace and Artifact Manifest

Every numerical claim in the main paper traces to a result file under `results/revision/` with a corresponding pre-specified script under `experiments/`. Table 17 summarises the trace; the full mapping is shipped as `SOURCE_TRACE.md` in the artifact and reproduced here for the reviewer’s convenience.

Table 17: Audit trace from claim to source file.

Audit	Source results files
Matched-context (§3 backbone)	<code>results/revision/fix_01/</code>
Scaled audit (§2)	<code>results/revision/fix_02/</code>
Metric fragility (Mistral)	<code>results/revision/fix_03/</code> , <code>standardized_scorer_fragility.csv</code>
Threshold transfer (§11)	<code>results/revision/fix_04/</code>
Noise (§10)	<code>results/revision/fix_05/</code>
Cost-aware head-to-head	<code>results/revision/fix_06/</code> , <code>h2h_summary_with_ci.csv</code> , <code>fix_11/</code>
Span/control (§3)	<code>results/revision/fix_12/</code>
Stronger retrievers (§4)	<code>results/revision/fix_13/</code> , <code>results/multi_retriever/</code>
Second generator (§5)	<code>results/revision/fix_14/</code>
Long-form stress (§8)	<code>results/revision/fix_15/</code>
Human calibration $n = 99$ (§12)	<code>human_eval_final/n99_calibration/</code>
Targeted disagreement evaluation $n = 100$ (§13)	<code>human_eval_final/n100_disagreement/</code>

Reproducibility. An anonymized artifact package containing code, per-query CSVs, matched context pairs, source tables, scripts, and documentation is provided for review. Public package, benchmark, and permanent archive links are omitted during double-blind review and will be released upon acceptance.

15 Resume and Compute Notes

The local scripts are resumable through their CSV outputs. The only script likely to benefit from a free-tier GPU notebook is the second-generator replication if local Ollama is slow: run the same command on a free GPU-backed notebook after pulling Qwen2.5 and Mistral into Ollama, then download `data/revision/fix_14/` and `results/revision/fix_14/`. None of the audit experiments require paid APIs; all scorer rescoring is local DeBERTa, `roberta-large-mnli`, and a Mistral-via-Ollama RAGAS-style judge.

16 Positioning Against Recent RAG Evaluation Work

The main paper compresses the related-work positioning into a single paragraph in the “Related Work, Continued” section. The detailed mapping is reproduced here for reviewer convenience.

Table 18: Positioning against recent RAG evaluation work.

Work	Primary contribution	ControlledRAG relation
ARES [Saad-Falcon et al., 2024]	trained judges for context relevance, answer faithfulness, and answer relevance with PPI calibration	complementary evaluator; ControlledRAG asks whether claims survive scorer/retriever/generator choices
RAGChecker [Ru et al., 2024]	fine-grained retrieval and generation diagnostics for RAG systems	complementary metric framework; ControlledRAG is the disclosure layer around such diagnostics
RAGTruth [Niu et al., 2024]	manually annotated hallucination corpus with word-level labels	complementary benchmark; ControlledRAG asks how benchmark conclusions transport across audit axes
MIRAGE [Park et al., 2025]	RAG benchmark with component-specific metrics and adaptability probes	complementary benchmark; ControlledRAG emphasizes reporting hidden degrees of freedom
FaithJudge [Tamber et al., 2025]	LLM-as-judge faithfulness benchmark and evolving leaderboard	complementary scorer/leaderboard; ControlledRAG requires scorer calibration and cross-scorer disclosure
RAGAS / TRUE / SummaC [Es et al., 2024, Honovich et al., 2022, Laban et al., 2022]	reference-free or NLI-style factual-consistency metrics	metrics used as audit inputs, not treated as reference standards

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